Aveon — Client Deployment Guide

Biometric Face-Reader Attendance Platform • FastAPI + React + SQL Server • Windows

Deploy the **AveonClientPackage** to a client server. Follow the phases in order. Commands marked *(admin)* require an elevated PowerShell (Start → PowerShell → Run as administrator).

Phase 1 Prerequisites (install once, as Administrator)

1. **Python 3.10+** — install and tick “Add to PATH”.
2. **Node.js 18+** — install (adds to PATH).
3. **SQL Server** (Express is fine) + **ODBC Driver 18 for SQL Server**.
4. **For remote devices on port 80** (skip if devices use port 8000 on the LAN): IIS enabled, plus **URL Rewrite 2.1** and **ARR 3.0** (links in `deployment\iis\README.md`).
5. **For LAN SDK features** (Test Connection / Push-via-SDK): register `zkemkeeper.dll` with `regsvr32`.

Phase 2 Copy & Database

6. Extract **AveonClientPackage.zip** to `C:\Aveon`.
7. In **SSMS**, create a database — e.g. `AIFace`.
8. Open `C:\Aveon\database\schema.sql` and run it **once** against that DB.

■ It drops and recreates tables — never re-run it on a populated database.

Phase 3 Configure

9. Edit `C:\Aveon\backend\.env`:
   - `SQLSERVER_HOST / DATABASE / USER / PASSWORD`
   - `SECRET_KEY` → a strong random string
   - `ADMIN_PASSWORD` → change from `admin123`
10. Leave `C:\Aveon\frontend\.env` as-is (`VITE_API_BASE_URL=` empty = same-origin; this is correct).

Phase 4 Install & Start

11. Double-click **C:\Aveon\install-and-start.cmd** — creates the Python venv, installs dependencies, builds the frontend, and starts the backend (port 8000) and frontend.
12. Verify: open `http://localhost:8000/health` → `{"status":"ok"}`.

Phase 5 Run backend as a service (survives reboot) (admin)

13. In an Administrator PowerShell, register the auto-start task:

```
$py='C:\Aveon\backend\.venv\Scripts\python.exe'; $dir='C:\Aveon\backend'
$a=New-ScheduledTaskAction -Execute $py -Argument '-m uvicorn app.main:app --host 127.0.0.1
--port 8000' -WorkingDirectory $dir
$t=New-ScheduledTaskTrigger -AtStartup
$p=New-ScheduledTaskPrincipal -UserId 'SYSTEM' -LogonType ServiceAccount -RunLevel Highest
$s=New-ScheduledTaskSettingsSet -RestartCount 3 -RestartInterval (New-TimeSpan -Minutes 1)
-ExecutionTimeLimit ([TimeSpan]::Zero)
Register-ScheduledTask -TaskName 'AveonBackend' -Action $a -Trigger $t -Principal $p
-Settings $s -Force
Start-ScheduledTask -TaskName 'AveonBackend'
```

Phase 6 Expose on port 80 for remote devices (IIS proxy) (admin)

Skip if devices will use port 8000 directly on the LAN.

14. In Administrator PowerShell:

```
cd C:\Aveon\deployment\iis
.\setup-iis-proxy.ps1 -HostName devices.yourdomain.com
```

15. Network: public **DNS A-record** for that hostname → server's public IP; forward inbound **TCP 80** to the server; allow port 80 in the firewall.
16. Verify from another machine: `http://devices.yourdomain.com/health` → ok.

Phase 7 Register devices & point them at the server

17. Open the admin site, log in as `admin`.
18. **Devices** page → add each device by its **exact serial number**.

■ Unregistered serials are rejected and their punches are lost. Register the serial BEFORE pointing the device.

19. On each device: **Comm.** → **Cloud Server / ADMS** → Server = hostname (or LAN IP), **Port = 80** (or 8000 if LAN-direct), HTTPS off. Reboot.
20. Confirm **Last Ping** updates on the Devices page within ~30s.

Phase 8 Nightly attendance recalculation

21. Create a Task Scheduler daily job:

Program	`C:\Aveon\backend\.venv\Scripts\python.exe`
Arguments	`scripts\recalc_attendance.py --days 35`
Start in	`C:\Aveon\backend`
Trigger	Daily, e.g. 01:00

Add `--export` if you configured `ATTENDANCE_EXPORT_*` in .env for payroll CSV/FTP.

Phase 9 Go-live checklist

- ■ http://localhost:8000/health returns ok (backend service running)
- ■ http://<hostname>/health returns ok (IIS proxy, if used)
- ■ Admin UI loads; logged in; admin password changed from default
- ■ Each device serial registered; device Cloud Server set
- ■ Device Last Ping updating; a test punch appears in Attendance Logs
- ■ Nightly recalc task scheduled; a manual run succeeds
- ■ Rollback path known if a hostname was migrated (deployment\iis\rollback-hostname.ps1)

Two critical reminders: **(1)** register a device's serial before pointing it at the server (Phase 7.18) — skipping this silently loses attendance; **(2)** change the admin password, since the login is internet-reachable through the proxy. Full details, plus hostname migration and FTP export, are in `DEPLOYMENT-STEPS.md` inside the package.